



Operating Instructions

Diesel engine
12V4000Gx3x
16V4000Gx3x

M015710/11E

Engine model	Frequency	Application group *
12V4000G23	50 Hz	3B, 3D, 3E, 3F, 3G
12V4000G23R	50 Hz	3B
12V4000G43	60 Hz	3A, 3B, 3D, 3F, 3G
12V4000G63	50 Hz	3A, 3B, 3D, 3E, 3F, 3G
12V4000G83	60 Hz	3A, 3B, 3D, 3F, 3G
16V4000G23	50 Hz	3B, 3D, 3E, 3F, 3G
16V4000G43	60 Hz	3A, 3B, 3D, 3F, 3G
16V4000G63	50 Hz	3A, 3B, 3D, 3E, 3F, 3G
16V4000G83	60 Hz	3A, 3B, 3D, 3F, 3G
16V4000G83L	60 Hz	3D
* 3A Heavy duty, unrestricted, ICXN 3B Continuous service, variable load, ICXN 3D Emergency standby power, IFN 3E Emergency service, ICXN 3F Heavy duty for DCP, unrestricted, ICXN 3G Heavy duty, intermittent, ICXN		

Table 1: Applicability

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California Proposition 65

This warning applies only to the State of California, USA, in compliance with Title 27 California Code of Regulations, Article 6, Clear and Reasonable Warnings.

 **WARNING:** Breathing diesel engine exhaust exposes you to chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

- Always start and operate the engine in a well-ventilated area.
- If in an enclosed area, vent the exhaust to the outside.
- Do not modify or tamper with the exhaust system.
- Do not idle the engine except as necessary.

For more information go to www.P65warnings.ca.gov/diesel.

(→ www.P65warnings.ca.gov/diesel)

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1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper and safe operation of your product from the manufacturer Rolls-Royce Solutions.

2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at: <http://www.mtu-solutions.com>

2.2 Intended use of gendrive engines

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the admissible operating parameters in accordance with the (→ technical sales documents – product data – TVU data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With preservation approved by the manufacturer in accordance with the (→ Preservation and Represervation Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (also applies to engine control/parameters)
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- In compliance with the maintenance work and intervals specified in the (→ Maintenance Schedule) throughout the useful life of the product
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques

The product shall not be operated in explosive atmospheres unless it fulfills the conditions for such use and approval has been granted.

The section below applies to applications deployed in regions under the jurisdiction of the EPA.

- Use of EPA Tier 2 and Tier 3 certified genset engines.
- EPA Tier 2 and Tier 3 certified genset engines may only be used in generator sets for stationary emergency applications. Non-emergency operation is limited to 100 hrs per year.

The engines may be used in fixed speed engine-generator sets only. Permanent limitation to a max. engine power of less than 560 kW (e.g. by means of correspondingly low generator power) is not admissible.

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:

- <http://www.mtu-solutions.com>

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.